

Polarimetric Observable based Optical Remote Sensing Systems for Heterogeneous Layered Scattering Environments

Haojie Ding^a, Menglei Ding^b, Tianyi Lv^b, Xiaopeng Gao^a, Xixun Sun^a, Dekui Li^{a,*}, Zhongyi Guo^{a,*}

^a School of Computer and Information, Hefei University of Technology, Hefei, 230009, China

^b 713th Research Institute, China State Shipbuilding Corporation Limited (CSSC), Zhengzhou, 450015, China

ARTICLE INFO

Keywords:

Polarization characteristics
Target detection
Stokes parameters
Monte Carlo simulation
Light scattering

ABSTRACT

The high-performance remote sensing imaging is always prevented by scattering environments, especially the scattering environments that are heterogeneous along transmitting direction of light, as the evolution of optical field becomes more complicated. To resolve this problem, this paper constructs a polarimetric observable based optical remote sensing system, which is able to represent heterogeneous scattering environments and moreover, identify the studying targets. On the one hand, the Monte Carlo (MC) algorithm and photon-tracking framework are employed to ensure that the proposed model can mimic actual heterogeneous remote sensing environments well. On the other hand, the proposed model takes advantage of polarimetric observables to identify targets by highlighting their physical meaning. The relevant results demonstrate that the proposed method is capable of identifying studying targets at the optical thickness of 7L that is unreachable by other optical methods. From this perspective, this paper provides a practical method to extend the detection distance of remote sensing and improve the detection ability of optical systems, which promotes the development of high-performance target detection in complex remote sensing environments.

1. Introduction

The remote sensing technology [1–3] has become a crucial tool to monitor climate change, study the Earth's environments, and importantly, detect targets. However, complex atmospheric environments always challenge high-performance target detection [4] in remote sensing systems. For instance, aerosols [5], clouds [6], moisture and other factors [7] dispersed in the atmosphere environments significantly affect the propagation path and intensity of light, resulting in the relevant imaging results of targets blurred. Moreover, the scattering environment is usually heterogeneous along the transmission path of light, which modulates the optical field in an extremely complex way. In such cases, traditional optical imaging techniques cannot obtain sufficient information of studying targets, failing to identify targets. For this reason, it is a high requirement to explore new detection methods that can provide information of studying targets effectively, so as to improve the accuracy and reliability of target detection in complex atmospheric circumstances, especially in heterogeneous scattering environments [8–12].

Polarization, as a fundamental property of light, provides target

information from a new perspective. In particular, the state of polarization (SoP) of incident light changes after interacting with a target [13, 14]. Therefore, through analyzing the changes in SoP of incidence, the polarization characterizations of interacted targets can be derived. It is worth noting that these polarization characteristics can provide information that cannot be obtained by traditional optical detection methods [15–17]. For instance, by analyzing the SoP of light, especially the degree of polarization (DoP) and angle of polarization (AoP), the target can be recognized effectively [14,18–20]. Furthermore, polarization characteristics help to distinguish between camouflaged targets and natural backgrounds, improving military target recognition [21], as well as enhancing disaster monitoring [22].

So far, the most advanced tool to analyze the polarization characterizations is using a polarimeter to measure the Muller matrix (MM). MM is a 4×4 matrix that contains all polarization characterizations. Besides, through applying decomposition methods to MM [23], it is possible to derive a series of polarimetric observables that are able to highlight corresponding polarization characterizations. Importantly, a set of polarimetric observables has demonstrated their superiority in

* Corresponding authors.

E-mail addresses: lidekuifut@163.com (D. Li), guozhongyi@hfut.edu.cn (Z. Guo).

<https://doi.org/10.1016/j.optlaseng.2025.108916>

Received 3 January 2025; Received in revised form 11 February 2025; Accepted 23 February 2025

Available online 6 March 2025

0143-8166/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

improving imaging quality in scattering environments [24–26]. For instance, the Indices of polarization purity (IPPs) [27,28] have been employed in biological imaging and coated target detection, and have proved their ability of eliminating the effects of scattering media.

In consideration of the factors stated above, this paper aims to achieve high-performance remote sensing in heterogeneous media by highlighting the advantages of polarimetric observables [29–31]. We first study the behavior of photons transmitting in heterogeneous atmosphere environments, based on the fact that imaging quality in atmosphere environments is affected by scattering and absorption [32–34]. Next, the photon evolution resulting from its interactions with various atmospheric particles is traced, which assists Monte Carlo(MC) algorithm in constructing a layered scattering media that is able to mimic actually heterogeneous scattering environments. Last but not least, the studying targets in different constructed scattering environments controlled with parameters measured in actual atmosphere environments are represented by polarimetric observables that enhance the imaging quality by highlighting polarization characterizations in different perspectives [35–37]. Combining the three steps, the polarimetric observable based remote sensing systems are constructed to resolve the target detection tasks in heterogeneous layered scattering environments.

The outline of this paper is arranged as follows: Section 2 presents the theory used in this paper, including the calculated method and corresponding physical meaning demonstrated in subsection 2.1, the scattering process shown in subsection 2.2, and the constructing layered model presented in section 2.3. Next, corresponding results in different scattering environments obtained by the proposed method, as well as its applicable boundary are presented in section 3. Finally, section 4 summarizes this paper.

2. Theory and Method

2.1. Polarimetric observables

As stated above, the physical property of targets is reflected in the change in SoP of the light interacting with the targets. That is to say, the physical property of targets can be obtained by analyzing the modification of polarization information of the light. To do so, in this section, the representation methods of polarization information of light and targets are presented.

At first, there is a polarimetric matrix, so-called Stokes vector (SV), which is widely used to represent the SoP of light. It is a 4×1 vector with the following form:

$$\mathbf{S} = [I, Q, U, V]^T, \quad (1)$$

as:

$$\mathbf{S}_{out} = \mathbf{M}\mathbf{S}_{in}, \quad (3)$$

where $\mathbf{M}$ in Eq.(3) is called a MM, which describes the modulation of polarization resulting from the interaction of light with samples. It has 16 elements with the following form:

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix}. \quad (4)$$

Since SV can represent all possible SoP of light, MM is able to represent all polarimetric characterizations contained in a sample. In this vein, MMs are used to analyze polarization characterizations of studying targets, such as overall polarizance (P_p) of a sample and degree of spherical purity (P_s), which can be respectively calculated by a normalized MM in term of m_{11} , with the following formalize:

$$P_p = \sqrt{\left(\sum_{i=2}^4 m_{i1}^2 + \sum_{j=2}^4 m_{1j}^2 \right) / 2}, \quad (5)$$

$$P_s = \frac{\|\mathbf{m}\|}{\sqrt{3}}, \quad (6)$$

where $\mathbf{m}$ is a 3×3 submatrix of the normalized $\mathbf{M}$. In addition to the polarimetric observables shown above, the determinant and trace of $\mathbf{M}$ can also be directly calculated from MM, shown as follows:

$$DET = |\mathbf{M}|, \quad (7)$$

$$TRA = m_{11} + m_{22} + m_{33} + m_{44}, \quad (8)$$

where, DET is the determinant of $\mathbf{M}$ and TRA is the trace of $\mathbf{M}$.

In addition to such polarimetric observables that can be directly obtained by MM, more polarimetric characterizations are codified in MM, and cannot be directly derived from the elements in MM, such as retardance and depolarization. In this regard, there are a series of decomposition methods for MM to obtain more polarimetric observables to represent related polarimetric characterizations, such as parallel decomposition [38] and Lu-Chipman decomposition [39].

In particular, the parallel decomposition considers that a depolarizer can be regarded as an incoherent summation of up to four polarization pure systems, and more importantly, the weights of such polarization pure system equal the eigenvalues of $\mathbf{H}$ matrix of MM. The relation between $\mathbf{H}$ and MM is illustrated as follows [40]:

$$\mathbf{H} = \frac{1}{4} \begin{bmatrix} m_{11} + m_{12} + m_{21} + m_{22} & m_{13} + m_{23} + i(m_{14} + m_{24}) & m_{31} + m_{32} - i(m_{41} + m_{42}) & m_{33} + m_{44} + i(m_{34} - m_{43}) \\ m_{13} + m_{23} - i(m_{14} + m_{24}) & m_{11} - m_{12} + m_{21} - m_{22} & m_{33} - m_{44} - i(m_{34} + m_{43}) & m_{31} - m_{32} - i(m_{41} - m_{42}) \\ m_{31} + m_{32} + i(m_{41} + m_{42}) & m_{33} - m_{44} + i(m_{34} + m_{43}) & m_{11} + m_{12} - m_{21} - m_{22} & m_{13} - m_{23} + i(m_{14} - m_{24}) \\ m_{33} + m_{44} - i(m_{34} - m_{43}) & m_{31} - m_{32} + i(m_{41} - m_{42}) & m_{13} - m_{23} - i(m_{14} - m_{24}) & m_{11} - m_{12} - m_{21} + m_{22} \end{bmatrix} \quad (9)$$

where I is the total intensity of the light, Q , U , and V are the discrepancy between 0° and 90° components, 45° and 135° components, as well as right-circle and left-circle components, respectively. Importantly, according to the SV, a polarimetric observable, DoP, representing the purity of light, can be calculated by:

$$\text{DoP} = \sqrt{Q^2 + U^2 + V^2} / I. \quad (2)$$

The SV will alter when incidence interacts with a target. The relation between incident polarization and emergent polarization is represented

Subsequently, according to the weight of such polarization pure systems, a series of polarimetric observables are derived. For instance, IPPs consisting of three polarimetric observables P_1 , P_2 , P_3 , are derived from the covariance matrix $\mathbf{H}$ (weight of polarization pure systems):

$$P_1 = \frac{\lambda_0 - \lambda_1}{tr\mathbf{H}}, \quad (10)$$

$$P_2 = \frac{(\lambda_0 - \lambda_2) + (\lambda_1 - \lambda_2)}{tr\mathbf{H}}, \quad (11)$$

$$P_3 = \frac{(\lambda_0 - \lambda_3) + (\lambda_1 - \lambda_3) + (\lambda_2 - \lambda_3)}{tr\mathbf{H}}. \quad (12)$$

Here, $\lambda_0, \lambda_1, \lambda_2$ and λ_3 are the four eigenvalues of matrix $\mathbf{H}$. They can represent numerous fundamental properties of a matrix, such as stability, transformation behavior, or spectral characteristics.

By the combination of elements in IPPs, a polarimetric observable, called depolarization index (P_Δ), representing the overall depolarization of samples can be derived [35,40]:

$$P_\Delta = \frac{1}{\sqrt{3}} \sqrt{2P_1^2 + \frac{2}{3}P_2^2 + \frac{1}{3}P_3^2}. \quad (13)$$

In comparison with the polarimetric observable P_Δ , IPPs are able to more comprehensively describe depolarization characterizations of studying samples. Therefore, IPPs are widely employed to distinguish between depolarizers with different origins [35,38].

2.2. Theory of scattering process

According to the ratio of wavelength to the size of samples, the scattering divides to three categories (Rayleigh scattering, Mie scattering and Geometric scattering), where the Mie scattering is widely used to represent the interaction between light and particles in foggy and underwater environments. As stated above, the modification of SoP, resulting from the interaction between light and samples, can be represented by MM. In the case of Mie scattering, the single scattering can be represented by the MM that relies on the scattering angle in terms of the direction of incident light [41], and the function of MM with scattering angle is shown as follows:

$$\mathbf{M}(\theta) = \begin{bmatrix} \mathbf{m}_{11}(\theta) & \mathbf{m}_{12}(\theta) & 0 & 0 \\ \mathbf{m}_{21}(\theta) & \mathbf{m}_{22}(\theta) & 0 & 0 \\ 0 & 0 & \mathbf{m}_{33}(\theta) & \mathbf{m}_{34}(\theta) \\ 0 & 0 & \mathbf{m}_{43}(\theta) & \mathbf{m}_{44}(\theta) \end{bmatrix}, \quad (14)$$

where θ refers to the scattering angle. By employing the MM shown in Eq.(14), and the relation between incidence and output light shown in Eq. (3), the SoP of scattering light can be calculated.

However, regarding the light propagating in media, its SoP is defined within the coordinate system in terms of Z, called a global coordinate system. In contrast, the light interacting with particles has to be defined in the coordinate system with respect to the particle, called a local coordinate system [42]. In this regard, it is inevitable to align such two coordinate systems. Herein, a rotation matrix is used to accomplish this task. It is able to alter the perspective of the polarized light, so as to alter the coordinate system of light. For this reason, the rotation matrix has the same form as linear retarders, and the corresponding MM is a function of the rotation angle:

$$\mathbf{R}(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(2\alpha) & \sin(2\alpha) & 0 \\ 0 & -\sin(2\alpha) & \cos(2\alpha) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (15)$$

where α refers to the rotation angle. For a single scattering process, the light first needs to be transformed from the global coordinate system to the local coordinate system, then the light interacts with a particle, and finally transformed from the local coordinate system to the global system. Therefore, the whole process of a single scattering can be expressed as:

$$\mathbf{M} = \mathbf{R}(\gamma)\mathbf{M}(\theta)\mathbf{R}(\beta), \quad (16)$$

where γ and β are the rotation angle, and θ is the scattering angle. θ and γ can be obtained by the rejection method [43]. The corresponding angles are presented in Fig. 1. Note that in addition to the angles used in Eq. (16), Fig. 1 also presents ϕ referring to the difference in azimuth angle between incident light and scattered light, which will be used to

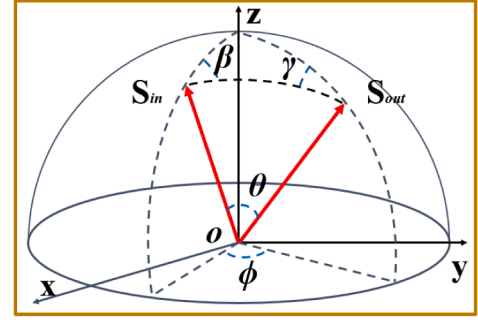


Fig. 1. Schematic of the single scattering event.

determine the next scattering position presented in section 2.3

In fact, photons usually continually undergo a series of scattering events in real scattering environments before they get captured. Under this scenario, the SoP of captured light can be regarded as being modulated by particles time by time. Accordingly, the SoP of captured light after multiply scattering can be calculated by the following equation:

$$\mathbf{S}_{out} = \prod_{k=0}^N \mathbf{R}(\gamma)\mathbf{M}(\theta)\mathbf{R}(\beta)\mathbf{S}_{in} = \mathbf{M}\mathbf{S}_{in}, \quad (17)$$

where N denotes the number of scattering events when light transmits in studying scattering media.

2.3. The Layered model and Monte Carlo algorithm

Taking the content shown in the last subsection, both SoP and scattering direction of photons can be determined after interacting with a particle [43]. In this vein, if the scattering distance can be determined, the next scattering position of the scattering photon can be calculated based both on the scattering direction and distance. In fact, the scattering distance of photon relies on the scattering media, forming a function of extinction coefficient with the following formula:

$$l = -\frac{\ln \xi}{u_e}, \quad (18)$$

where, ξ is a random number between 0 and 1, and u_e is the extinction coefficient. Through random sampling ξ , the scattering distance can be determined for each scattering event.

Summarizing, the process of light transmitting in scattering media can be regarded as the continual interaction between photons and particles. Moreover, the continual interactions consist of a series of single scattering events in which the scattering direction is obtained rejection method and the scattering distance is assigned by Eq.(18). Based on such content shown above, the behavior of photons transmitting in the homogeneous scattering media is able to be reproduced. Beyond homogeneous scattering media, this paper tries to explore how photons behave in heterogeneous scattering media, and construct a layered

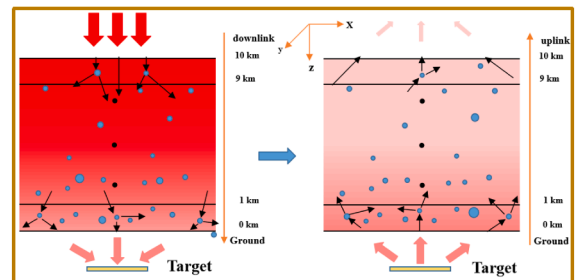


Fig. 2. The layered heterogeneous atmosphere environment.

scattering media to mimic the actual heterogeneous remote sensing environments.

To do this, the length of the scattering media is set a 10 km for 10 layers, as shown in Fig. 2, with different controlling parameters, i.e. the radius and density of particles. In particular, each layer is set as 1km, and moreover, the particle number density (PND) describing each layer is obtained through measuring the real atmosphere environment at Hefei, Anhui, China (N31°, E117°) [44], as shown in Tab. 1

Notably, as can be seen from Tab. 1, the PND is significantly thin, which means that the scattering events mainly occur in the range of 10km below the troposphere. It demonstrates it is reasonable to set the longest distance of our model as 10km. Actually, the behavior of photons in layered scattering environments is more complex than in uniform scattering environments, as the parameters are different from layer to layer. In particular, even though varied parameters among different layers do not affect the scattering direction determined by the rejection method, such parameters have significant affections on the scattering distance, as the u_e is varied over the layers in which the photon transmits. Moreover, in the constructed conditions, the particle size distributions are also assigned into ten categories that respectively correspond to the ten layers. They are 0.02-0.05 μm , 0.05-0.1 μm , 0.1-0.125 μm , 0.125-0.25 μm , 0.25-0.5 μm , 0.5-0.8 μm , 0.8-1.0 μm , 1.0-1.25 μm , 1.25-5 μm , and 5-100 μm from top to bottom. By altering the proportion of particles within different size ranges and incorporating the PND presented in Tab. 1, it is possible to effectively simulate atmospheric environments across various dimensions. In order to clarify this issue, a scattering event in layered scattering environments is present in Fig. 3.

Fig. 3 presents a scattering of photons in the layered scattering environments. Let us assume that the photon interacts with a particle at the $P_0(x_0, y_0, z_0)$ where the photon starts to transmit with the direction $d_0(u_{x0}, u_{y0}, u_{z0})$. For the uniform scattering media that have the same parameters over the whole media, it is easy to calculate the scattering distance using Eq. (18). However, the scattering distance in the heterogeneous cannot be calculated directly due to the changing parameters. In this regard, we propose to use the transmittance method to resolve the problem of the calculation of scattering distance.

In particular, we first employ the Lambert-Beer law to describe the energy conversion of photons:

$$\tau = \frac{I_{out}}{I_{in}} = e^{-lu_e} = e^{-\epsilon}, \quad (19)$$

where, τ is the transmittance, I_{out} and I_{in} indicate the scattered and incident light respectively, l is the distance, u_e is the extinction coefficient and ϵ is the optical thickness (OT). Regarding the atmosphere environment that is divided into numerous layers, the transmittance of the entire atmosphere environment can be regarded as the product of the transmittance of all layers:

$$\tau = \tau_{i+1} \tau_i \dots \tau_{i-j+2} \tau_{i-j+1}. \quad (20)$$

Applying the logarithm to both sides of Eq. (20), it becomes the

Table 1
The PND for 0-10km measured in Hefei, Anhui, China

	Clear sky	Overcast sky
Altitude(km)	PND(particle/cm ³)	PND(particle/cm ³)
0~1	2036.04	9730.02
1~2	890.55	3737.01
2~3	381.35	1259.05
3~4	173.40	461.15
4~5	104.54	167.54
5~6	76.62	76.62
6~7	61.18	61.18
7~8	58.84	58.84
8~9	59.44	59.44
9~10	57.47	57.47

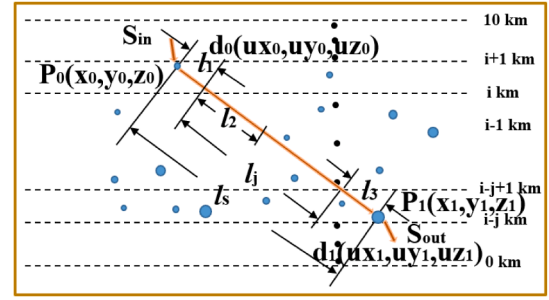


Fig. 3. A scattering event for photons in the layered atmosphere environments.

following form:

$$-\ln(\tau) = -\sum_{k=1}^j \ln(\tau_{i-k+1}) = u_e^{i+1} l_1 + \sum_{k=1}^{j-1} u_e^{i+1-k} l_2 + u_e^{i-j+1} l_3, \quad (21)$$

where l_1 represents the distance of the photon from the scattering starting point to the next layer, l_2 represents the scattering distance through the complete layer along the scattering direction in a single scattering event, and l_3 represents the distance of the photon from the final scattering position to the final layer along the scattering direction. Besides, combining Eq.(18) and Eq.(19), we find that τ is numerically equal to ξ . Based on the equation shown above, the total scattering distance can be written as:

$$l_s = l_1 + l_j + l_3 = l_1 + (j-1)l_2 + l_3, \quad (22)$$

where $l_j = (j-1)l_2$ denotes the scattering distance sum of all the complete layers traversed along the scattering direction in a single scattering, where j is obtained by Eq. (21). Obviously, by calculating l_1 , l_2 , l_3 , we can determine the total distance l_s , which allows us to determine the next scattering position of the photon. The calculation of l_1 , l_2 , l_3 can be categorized into three cases in terms of the azimuth angle ϕ , with the details are shown as follows:

Case1. $\cos(\phi) = 0$

This case indicates that the scattering direction of the photon is parallel to the current layer, which means that the subsequent scattering event will occur within the same layer. In this case, the path length can be directly determined using Eq. (18).

Case2. $\cos(\phi) > 0$

In this situation, photons are forward scattered within the medium. The remaining distance $l_1 = (z_0 + j)/u_{z0}$ can be calculated once the photon does not transition to the next layer from its current position. The propagation distance to the next layer is given by $l_2 = 1/u_{z0}$. If $l_{j+1} < l_1$, the next scattering event will still occur within the current layer. Otherwise, the photon will enter the next atmospheric layer. In this case, the expression for l_3 is given by

$$l_3 = \frac{-\ln(\xi) - l_1 u_e^{-(i+1)} - \left(l_2 \sum_{k=1}^{j-1} u_e^{i+1-k} \right) u(j-1)}{u_e^{i-j+1}}, \quad (23)$$

where $u(j-1)$ is the step function, j is added before each iteration. If $l_2 < l_3$, the next loop iteration starts. Otherwise, the iteration ends and l_s can be obtained by substituting l_1 , l_2 , l_3 into Eq.(22).

Case3. $\cos(\phi) < 0$

In this case, photons are backscattered with $l_1 = (j+1-z_0)/u_{z0}$. The calculation method remains the same as in Case 2 in which l_1 , l_2 , l_3 can be obtained by substituting into Eq. (22). Notably, the values of l_1 , l_2 , l_3 are all negative in this scenario so the conditions for judgment are contrary to that of Case 2 under such circumstances. When the photon is

located on the ground, both forward and backscatter need to be considered, taking into account the different values of the u_c .

In consideration of such three cases stated above, the scattering distance is able to be calculated to determine the next scattering position with the assistance of the scattering angle. Moreover, the SoP of photons after scattering can be obtained using Eq. (16). Under this scenario, the whole process of light transmitting in layered scattering environments can be traced. It is worth noting that in the constructed layered scattering environment, each photon has unique behavior that is determined by, on the one hand, the scattering angles assigned by the rejection method, and, on the other hand, the scattering distance that is calculated based on a random number. In this regard, the real scattering condition can be mimicked by the proposed method using the MC algorithm with sufficient sampling, and moreover, more sampling number (photon number), closer the constructed environment to the real one [37]. Herein, in consideration of the balance between required calculation sources and accuracy of results, the number of photons is set as 10^8 .

3. Results and analysis of the simulation

3.1. The imaging effect of different polarization parameters in the same atmospheric condition

Based on the real environment where the PND are measured by experiment, as shown in Fig. 2, a remote sensing model has been constructed in the last section. In this section, leveraging the constructed model, we study the imaging performance of a number of widespread polarimetric observables, and explore how to most effectively eliminate the scattering effects by analyzing the physical meaning behind such polarimetric observables. To do so, firstly, lights of four kinds SoP (corresponding SV are $[1,0,0,0]^T$, $[1,1,0,0]^T$, $[1,0,1,0]^T$, $[1,0,0,1]^T$), with a wavelength of 632.8nm, are launched into the scattering media (here are the constructed remote sensing environments). Regarding each radiation, the photon will undergo a series of scattering events before arriving at studied targets, and then transmit back with a series of scattering events to be captured (see Fig. 2). By repeating this process for the four kinds of light, the MM of the whole system can be obtained. Herein, the size of the studying target is $2m \times 2m$, consisting of three parts with different polarization characterizations. (see Fig. 4). In particular, the first part is made of steel shaped as the English letters "HFUT", the second part is made of wood shaped as a circle with a radius of 1m, and the last part is a $2m \times 2m$ square stone. The schematic of the target is illustrated as follows:

Note that all three parts of the targets are the depolarizers without diattenuation and retardance. The corresponding studies have shown empirically that, regarding these targets, their MMs have the following form [45–46]:

$$\mathbf{M}_{\text{Object}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & m_{22} & 0 & 0 \\ 0 & 0 & m_{22} & 0 \\ 0 & 0 & 0 & m_{44} \end{bmatrix}. \quad (24)$$



Fig. 4. The structure of the studying target. The first layer is steel with reflectance of 0.9, the second layer is wood with reflectance of 0.2, and the third layer is stone with reflectance of 0.7

Table 2
Mueller matrix elements

Object	m_{22}	m_{44}
Steel	0.975	0.99
Wood	0.215	0.16
Stone	0.55	0.45

It is easy to know from Eq. (24) that, m_{22} and m_{44} , these two elements co-determine the polarization characterizations of studying targets. Regarding the target studied here, the m_{22} and m_{44} of such three parts are 0.975, 0.215, 0.55 and 0.99, 0.16, 0.45. The details of elements in MM for studying samples are presented in Tab. 2.

The corresponding values of MM element for different parts indicate that the three studying parts have unique polarization characteristics. As a result, the imaging of such parts with different polarimetric observables will have unique performance. Before presenting the imaging of polarimetric observables, the OT has been calculated by Eq. (19) which describes the relation between incidence with intensity and scattering light captured by a $0.1m \times 0.1m$ area. Regarding this remote sensing environment, the OT is 0.819.

The imaging results of polarimetric observables introduced in section 2 are obtained by decomposing corresponding MM pixel by pixel, and the imaging results are illustrated in Fig. 5. There are ten polarimetric observables shown in Fig. 5. In particular, the polarimetric observable I refers to the intensity of scattering light, representing the reflection effect of the target, polarimetric observables DoP , P_1 , P_2 , P_3 , and P_Δ represent the depolarization characterizations, polarization observable P_p represent diattenuation, P_s implies retardance, DET is determinate, and the TRA represents the trace of MM. It is easy to see from Fig. 5 that the image of polarimetric observables I cannot distinguish between different parts of the studying target. It is understandable because before being captured by the detector, the photons underwent a series of scattering events that attenuate the intensity and randomize the direction of the photons, leading to the imaging of polarimetric observables being covered by something like a "fog". In addition to the polarimetric observable I , the polarimetric observable P_p cannot represent any information about targets. This phenomenon depends on the polarimetric characterizations of targets and scattering media. Besides such two polarimetric observables, other polarimetric observable images illustrate target information more or less. In particular, the corresponding images of polarimetric observables describing depolarization (DoP , P_1 , P_2 , P_3 , and P_Δ) illustrate the outline of different target parts, making it possible to distinguish between studying target parts made of different materials. Moreover, the other two polarimetric observable images, DET and TRA , contain an amount of target information, from which different target parts can be distinguished as well.

In summary, in comparison with intensity images, the polarimetric observable images, except polarization observable P_p , are able to reveal target information behind scattering media, leading that the targets with different polarimetric characterizations are capable of being identified

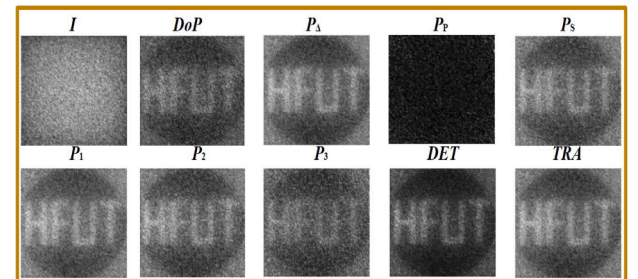


Fig. 5. The polarimetric observables at 0-10km latitude in a clear-sky environment.

through taking advantage of polarimetric observables. However, it is worth noting that polarimetric observables can highlight target information, but in different ways. For instance, the wood area in the *DET* polarimetric observable is significantly darker than that in other polarimetric observable images.

3.2. The difference in imaging effect in different weather

To extend the applicable range of the proposed model, so as to approximate real-world conditions, the case of overcast atmospheric environment is also studied. To construct an overcast environment, the parameter PND in Tab. 1 describing a overcast sky are regarded as a baseline. Following the same process of obtaining polarimetric observable images in clear-sky environments, corresponding polarimetric observable images are calculated and illustrated in the first row of Fig. 6. Note that the results obtained from clear-sky environment are illustrated in the second row of Fig. 6 for the comparison between different remote sensing environments.

To quantitatively represent the scattering environments of clear sky and overcast sky, the OT of such environments is calculated. That is, the OT of the clear sky environment is 0.819, which is significantly lower than that of the overcast environment with 4.571. As can be seen from Fig. 6, the results of the overcast sky environment are acceptable. Even though the corresponding results are not as clear as the results from the clear sky environment, the different parts of the target remain identifiable. It demonstrates that the proposed model is able to distinguish between targets, with highlighting the effort of polarimetric observables.

3.3. The influence of particle number density on imaging effect

As demonstrated above, the proposed model is capable of resolving the problem of target detection and analysis in remote sensing environments including clear and overcast sky environments. This subsection continually investigates the effects of the sky environment, controlled by the PND, on imaging performance so as to explore the applicable boundaries of the proposed method. To do so, different the PND are selected to construct a series of sky environments. Note that in this subsection, the OT is selected to quantitatively represent the complexity of sky environments, as OT is a widespread indicator of scattering environments. In particular, the OTs are assigned as 4, 5, 6, and 7, regarding which the corresponding results are obtained and illustrated in the following figure. The first row in Fig. 7 shows the reference image, while the subsequent rows display the results of different values of OT. Note that the reference images are generated in an ideal condition where no scattering medium exists between the detector and studying targets. Accordingly, the images present the real targets without any distortion resulting from scattering media.

The reason, why the OT is assigned from 4 to 7, is that, on the one hand, the last section has illustrated the imaging quality is acceptable at OT of 4.571. On the other hand, as shown in Fig. 7, the images are totally damaged at OT of 7 for all polarimetric observables. Moreover, through browsing the results at OT from 4 to 7, it is easy to see that all polarimetric observables can present the target well at OT of 5, even though the quality of images at OT of 4 is better than those at OT of 5. It is

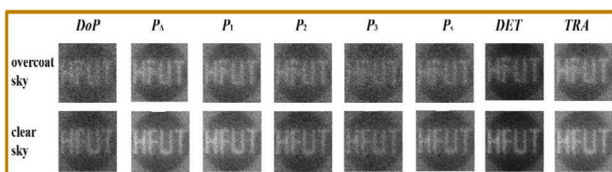


Fig. 6. The polarimetric observables at 0-10km latitude in a cloudy sky and clear sky.

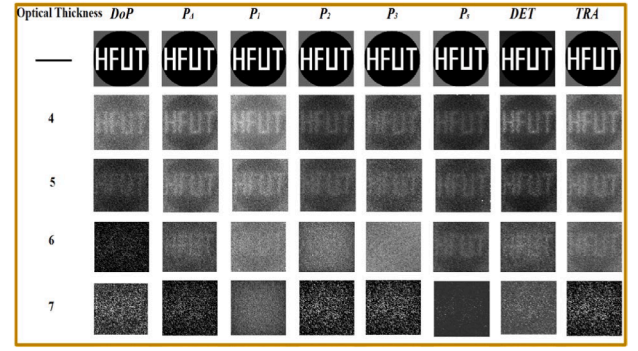


Fig. 7. The polarimetric observables at 0-10km latitude in the environments with different OTs.

understandable considering that the increasing OT means more scattering events that will decrease the imaging quality. Under this scenario, with the further increase of OT, polarimetric observables become more blurred, some of which start to lose the ability of identifying targets.

In particular, when OT arrives at 7, the images of polarimetric observables *DoP*, *P₁*, *P₂*, and *P₃* are diminished whereas the polarimetric observables *P₄*, *DET*, and *TRA* are still able to distinguish different parts. Furthermore, through comparing such three polarimetric observables *P₄*, *DET*, and *TRA*, it is easy to see that the polarimetric observable *P₄* is better than the other two polarimetric observables. To confirm this statement, four widespread indicators: peak signal-to-noise ratio (*PSNR*), structural similarity index (*SSIM*), contrast (*CON*) and entropy (*ENT*) are selected to quantitatively evaluate the quality of the illustrated polarimetric observables. Note that we select such four indicators to measure image quality not randomly, but considering the physical characterization of these indicators. For instance, *PSNR* is commonly used to evaluate the quality of the reconstructed image through comparing it with the original one, providing a measure of the similarity between them.

$$PSNR = 10 \lg \left(\frac{MAX^2}{MSE} \right), \quad (25)$$

where *MAX* represents the maximum value over all pixels in an image and *MSE* indicates the mean squared error. It is easy to understand that higher values of *PSNR* reflect the better quality of the reconstructed image.

Unlike *PSNR*, *SSIM* evaluates the imaging quality through considering luminance, contrast, and structural information with the following form:

$$SSIM = \frac{(2\mu_1\mu_2 + C_a)(2\sigma_{12} + C_b)}{(\mu_1^2 + \mu_2^2 + C_a)(\sigma_1^2 + \sigma_2^2 + C_b)}, \quad (26)$$

where μ_1 and μ_2 represent the mean values of the reconstructed image and ground truth respectively, σ_1 and σ_2 represent the standard deviation of the reconstructed image and ground truth respectively, σ_{12} represents covariance between the reconstructed image and ground truth, as well as C_a and C_b are constants set as 6.5 and 58.5, respectively. The definition of *SSIM* shown in Eq. (26) allows it to evaluate image quality in a more comprehensive way.

In addition to these image indicators shown above, there are two indicators to measure the difference between reconstructed image and ground truth.

$$CON = \frac{Contrast_1}{Contrast_2} = \frac{\frac{max_1 - min_1}{max_1 + min_1}}{\frac{max_2 - min_2}{max_2 + min_2}}, \quad (27)$$

where *max* and *min* represent the maximum and minimum values over all pixels in an image.

Table 3

Comparison of imaging results in different PND.

OT		DoP	P_{Δ}	P_1	P_2	P_3	P_s	DET	TRA
4	PSNR↑	11.36	11.96	11.82	11.55	11.61	16.47	9.66	7.20
	SSIM↑	0.13	0.22	0.21	0.16	0.12	0.27	0.24	0.12
	CON↓	0.29	0.22	0.22	0.26	0.38	0.28	0.09	0.22
	ENT↓	4.88	4.62	4.61	4.82	5.17	4.26	3.62	5.23
5	PSNR↑	11.24	11.90	11.74	11.39	11.20	16.26	9.64	7.13
	SSIM↑	0.11	0.19	0.19	0.13	0.10	0.23	0.23	0.11
	CON↓	0.33	0.24	0.24	0.29	0.44	0.32	0.10	0.24
	ENT↓	5.06	4.72	4.71	4.93	5.31	4.39	3.70	5.32
6	PSNR↑	8.72	9.82	9.03	8.74	8.77	13.86	9.32	5.33
	SSIM↑	0.06	0.15	0.10	0.06	0.05	0.20	0.22	0.09
	CON↓	0.66	0.33	0.54	0.61	0.61	0.75	0.49	0.45
	ENT↓	2.54	3.71	2.80	2.59	2.60	2.77	3.11	3.37
7	PSNR↑	8.30	7.72	7.80	7.34	6.27	11.72	9.03	2.93
	SSIM↑	0.02	0.02	0.03	0.02	0.01	0.09	0.08	0.01
	CON↓	0.93	0.77	0.52	1.00	1.35	0.36	0.79	1.14
	ENT↓	2.42	2.58	2.15	2.70	2.43	2.13	2.16	2.43

Table 4

Error ratio of imaging results.

	DoP	P_{Δ}	P_1	P_2	P_3	P_s	DET	TRA
PSNR	0.45%	0.49%	0.67%	0.78%	0.53%	0.42%	0.92%	1.10%
SSIM	1.01%	4.43%	4.16%	2.13%	3.05%	2.19%	4.31%	2.33%
CON	4.21%	2.19%	3.77%	2.18%	3.33%	4.00%	4.17%	3.45%
ENT	3.93%	1.96%	2.33%	3.61%	3.32%	3.29%	3.31%	2.50%

ENT denotes the ratio between the entropy of two images, measuring the complexity and information variation between the reference and result images.

$$ENT = \frac{H_1}{H_2} = \frac{\sum_{i=1}^L p_1(i) \log_2 p_1(i)}{\sum_{i=1}^L p_2(i) \log_2 p_2(i)}, \quad (28)$$

where L is the dynamic range (set as 256 here), and $p(i)$ indicates the possibility of the gray level i appearing in an image.

The definition of such indicators demonstrates that they evaluate imaging quality in different ways. That means the imaging quality is able to be comprehensively evaluated from these indicators. Thus, such indicators are applied to the results shown in Fig. 7, to understand and study how the imaging quality changes with OT. The corresponding results of these calculations are summarized in Tab. 3.

Tab. 3 presents image quality implied in values of four indicators, i. e. PSNR, SSIM, CON and ENT. Note that such four indicators present image quality in different ways. In particular, the higher the value of the PSNR and SSIM, the higher the quality of images, which is marked by an upward arrow. Conversely, regarding the indicators marked by a downward arrow, such as CON and ENT, an increase in these indicators means decreased imaging quality. Accordingly, it is easy to see that with the increase of OT, the value of PSNR and SSIM get bigger while the CON and ENT get smaller. This phenomenon demonstrates that the imaging quality gets worse with increased OT, which is consistent with the statement illustrated in Fig. 7. Most importantly, the quantitative results shown in Tab. 3 demonstrate the superiority of the polarimetric observable P_s in improving imaging quality, especially for the results at OT of 7 (see the color of corresponding values).

To evaluate the robustness of the proposed model, we repeat the model four times in the condition of OT = 4, and choose the error ratio (ER) as an indicator, which can be calculated as follows:

$$ER = \frac{d}{MIN}, \quad (29)$$

where d is the difference between the maximum and minimum values of each result, and MIN is the minimum value of each result. Using Eq.

(29), the ER of PSNR, SSIM, CON and ENT values are calculated and shown in Tab. 4. The corresponding results in Tab. 4 indicate that the maximum of ER is smaller than 5% (4.43%). This small value of ER verifies that the proposed model has a satisfied robustness.

4. Conclusion

In conclusion, this paper constructs a layered remote sensing model, based on the parameters measured in actual environments, to study the polarimetric characterizations of targets by highlighting the advantage of polarimetric observables. In particular, taking advantage of the MC algorithm and photon tracing, a 10-layer remote sensing environment has been proposed in this paper, which provides a practical method to construct the scattering media that is heterogeneous along the transmission direction of light. Then, using the constructed remote sensing environment, the polarimetric observables are employed to study the targets through considering the unique polarimetric characterizations implied in such polarimetric observables. The relevant results demonstrate that the polarimetric observables are able to reveal target information at the OT where the intensity imaging is out of strength.

Notably, the required time of the proposed model (including the time to obtain corresponding MMs and the time to calculate polarimetric observables from such MMs), not only depends on the parameters controlling remote sensing conditions (including the length of scattering environment, PND and the total number of photons), but also highly depends on the performance of the used computation devices. Specifically, for the computation device used in this paper, it can accomplish the model in 10 hours for the overcoat sky scenario, and 5 hours for clear sky.

In this paper, we study the physical properties of targets in an active detection way, which means that an active source is required to radiate the studying targets. This, to some content, limits the application ranges of the proposed method for remote sensing. For this reason, it is of high requirement to consider the spontaneous radiation of targets, so as to resolve the remote sensing scenarios in which the active detection is unachievable.

Funding

This research was funded by the National Natural Science Foundation of China under Grant No. 61775050. The computational resources were provided by the HPC Platform of Hefei University of Technology.

Data availability

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Haojie Ding: Writing – review & editing, Visualization, Software, Methodology. **Menglei Ding:** Visualization. **Tianyi Lv:** Formal analysis. **Xiaopeng Gao:** Visualization, Software, Formal analysis. **Xixun Sun:** Visualization, Software, Formal analysis. **Dekui Li:** Writing – review & editing, Visualization, Supervision, Formal analysis. **Zhongyi Guo:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

References

- [1] Yan L, Sun H, Tang H, et al. The design and implementation for UAV polarization remote sensing system. In: 2009 IEEE International Geoscience and Remote Sensing Symposium; 2009. p. 547–50.
- [2] Mei L, Rozanov V, Ritter C, et al. Retrieval of Aerosol Optical Thickness in the Arctic Snow-Covered Regions Using Passive Remote Sensing: Impact of Aerosol Typing and Surface Reflection Model. *IEEE Transactions on Geoscience and Remote Sensing* 2020;58(7):5117–31.
- [3] Tao Q, Guo Z, Xu Q, et al. Retrieving the polarization information for satellite-to-ground light communication. *Journal of Optics* 2015;17(8):085701.
- [4] Afonin A V, Davydov V S, Gorbunov G G, et al. Ground-based spectrometric complex for studying the concentration of anthropogenic gases in the boundary layer of the atmosphere and refining the data of satellite measurements. *Journal of Optical Technology* 2004;71(11):780–4.
- [5] Sano I, Mukai S, Okada Y. Aerosols all over the ocean and land, IGARSS 2001. Scanning the Present and Resolving the Future. *Proceedings. In: IEEE 2001 International Geoscience and Remote Sensing Symposium (Cat. No.01CH37217). 7; 2001. p. 3102–4.*
- [6] Wang Y, Shang H, Letu H, et al. Impact of Orbital Characteristics and Viewing Geometry on the Retrieval of Cloud Properties From Multiangle Polarimetric Measurements. *IEEE Transactions on Geoscience and Remote Sensing* 2023;61:1–17.
- [7] Hu T, Shen F, Wang K, et al. Broad-Band Transmission Characteristics of Polarizations in Foggy Environments. *Atmosphere* 2019;10(6):342.
- [8] Fan X, Chen W, Lin B, et al. Improved polarization scattering imaging using local-global context polarization feature learning framework. *Optics and Lasers in Engineering* 2024;178:108194.
- [9] Fan X, Lin B, Guo K, et al. TSMPN-PSI: high-performance polarization scattering imaging based on three-stage multi-pipeline networks. *Optics Express* 2023;31(23):38097–113.
- [10] Lin B, Fan X, Peng P, et al. Dynamic polarization fusion network (DPFN) for imaging in different scattering systems. *Optics express* 2023;32(1):511–25.
- [11] Lin B, Fan X, Peng P, Li D, et al. Research progress of imaging through scattering media based on deep learning. *Chinese Journal of Quantum Electronics* 2022;39(6):880.
- [12] Yu W, Shah S A A, Li D, et al. Polarized computational ghost imaging in scattering system with half-cyclic sinusoidal patterns. *Optics & Laser Technology* 2024;169:110024.
- [13] Li X, Goudail F. Vortex retarder-based Stokes polarimeters: optimal data processing and autocalibration capability. *Optics letters* 2024;49(7):1696–9.
- [14] Li B, Chen Z, Lu L, et al. Cascaded frameworks in underwater optical image restoration. *Information Fusion* 2025;117:102809.
- [15] Popov A, Bortsova M. Detection of Metal Objects Against Mountain Ranges Using Polarization Transfer Functions. In: 2018 IEEE 17th International Conference on Mathematical Methods in Electromagnetic Theory (MMET); 2018. p. 290–3.
- [16] Li L, Li Z, Huang Y, et al. Simulation of the polarization pattern of skylight affected by mineral dust aerosol particles. In: 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS); 2016. p. 4093–5.
- [17] Fan X, Lin B, Guo Z. Infrared Polarization-Empowered Full-Time Road Detection via Lightweight Multi-Pathway Collaborative 2D/3D Convolutional Networks. *IEEE Transactions on Intelligent Transportation Systems* 2024;25(9):12762–75.
- [18] Tao Q, Sun Y, Shen F, et al. Active imaging with the aids of polarization retrieve in turbid media system. *Optics Communications* 2016;359:405–10.
- [19] Xu Q, Guo Z, Tao Q, et al. A novel method of retrieving the polarization qubits after being transmitted in turbid media. *Journal of Optics* 2015;17.
- [20] Liao R, Yong W, Ma H. A potential method to detect the orientation of the suspended rod-shaped particles using polarized-light scattering. *OCEANS 2015 - MTS/IEEE Washington*; 2015. p. 1–5.
- [21] Lavigne D A, Breton M, Pichette M, et al. Enhanced Military Target Discrimination using Active and Passive Polarimetric Imagery, IGARSS 2008. In: 2008 IEEE International Geoscience and Remote Sensing Symposium; 2008. p. 354–7.
- [22] Xie B, Pan H, Xiang Z, et al. Polarization-Based Water Hazards Detection for Autonomous Off-road Navigation. In: 2007 International Conference on Mechatronics and Automation; 2007. p. 1666–70.
- [23] Zhao Y, He W, Ren H, et al. Polarization Decoupling Imaging of Underwater Complex Targets Based on Mueller Matrix Decomposition. *IEEE Photonics Journal* 2022;14(5):1–6.
- [24] Shen F, Zhang B, Guo K, et al. The Depolarization Performances of the Polarized Light in Different Scattering Media Systems. *IEEE Photonics Journal* 2018;10(2):1–12.
- [25] Shen F, Wang K, Tao Q, et al. Polarization imaging performances based on different retrieving Mueller matrices. *Optik - International Journal for Light and Electron Optics* 2018;153:50–7.
- [26] Sharma B S M, Sujatha N. Assessment of skin fibrosis using Mueller matrix polarimetry. In: 2018 3rd International Conference on Microwave and Photonics (ICMAP); 2018. p. 1–2.
- [27] Li D, Guo K, Sun Y, et al. Depolarization Characteristics of Different Reflective Interfaces Indicated by Indices of Polarimetric Purity (IPPs). *Sensors* 2021;21.
- [28] Wang P, Li D, Wang X, et al. Analyzing Polarization Transmission Characteristics in Foggy Environments Based on the Indices of Polarimetric Purity. *IEEE Access* 2020;8:227703–9.
- [29] Kyselák M, Grenar D, Slavíček K. Smart Sensor System for Detecting Temperature Changes by Polarized Light. In: 2021 6th International Conference on Smart and Sustainable Technologies (SpliTech); 2021. p. 1–3.
- [30] Li X, Huang Q. Target detection for infrared polarization image in the background of desert. In: 2017 IEEE 9th International Conference on Communication Software and Networks (ICCSN); 2017. p. 1147–51.
- [31] Wang X, Hu T, Li D, et al. Performances of Polarization-Retrieve Imaging in Stratified Dispersion Media. *Remote Sensing* 2020;12:2895.
- [32] Guo Z, Wang X, Li D, et al. Advances on theory and application of polarization information propagation(Invited). *Infrared Laser Engineering*; 2020.
- [33] Li D, Xu C, Lin B, et al. Research progress of polarization purity index theory and application. *Infrared and Laser Engineering* 2022;51(3). 20210373-1-73-13.
- [34] Motahar T, Rahman R, Hossain R. A Simulation Study on Light Scattering Effect on Water-borne Bacteriophage Virus Using Mie Analysis. In: 2017 IEEE 17th International Conference on Bioinformatics and Bioengineering (BIBE); 2017. p. 255–9.
- [35] Li D, Ma D, Guo K, et al. Polarization characteristics motivating target detection in different polarization spaces. *Optics & Laser Technology* 2024;171:110430.
- [36] Li D, Xu C, Zhang M, et al. Measuring glucose concentration in a solution based on the indices of polarimetric purity. *Biomedical Optics Express* 2021;12(4):2447–59.
- [37] Li M, Lu P, Yu Z, et al. Vector Monte Carlo simulations on atmospheric scattering of polarization qubits. *Journal of the Optical Society of America A* 2013;30:448–54.
- [38] Gil J J. Structure of polarimetric purity of a Mueller matrix and sources of depolarization. *Optics Communications* 2016;368:165–73.
- [39] Lu S-Y, Chipman RA. Interpretation of Mueller matrices based on polar decomposition. *Journal of the Optical Society of America A* 1996;13(5):1106–13.
- [40] Tariq A, He H, Li P, et al. Purity-depolarization relations and the components of purity of a Mueller matrix. *Optics Express* 2019;27(16):22645–62.
- [41] Mie G J A d P. Beitrage zur Optik trueber Medien, speziell kolloidaler Metalloesungen. 25.
- [42] Rybicki GB. Radiative transfer. *Journal of Astrophysics and Astronomy* 1996;17(3):95–112.
- [43] Raković M J, Kattawar G W, Mehrübeoğlu M, et al. Light backscattering polarization patterns from turbid media: theory and experiment. *Applied Optics* 1999;38(15):3399–408.
- [44] Wu D, Zhou J, Liu D, et al. 12-year LIDAR Observations of Tropospheric Aerosol over Hefei (31.9°N, 117.2°E). *China. Journal of the Optical Society of Korea* 2011;15(1):90–5.
- [45] Sebastien B, Philippe C. Modeling and performances of a polarization active imager at $\lambda=806\text{nm}$. *Optical Engineering* 2000;39(10):2681–8.
- [46] Chun C S L, Sadjadi F A. Polarimetric laser radar target classification. *Optical letters* 2005;30(14):1806–8.